

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

Both the deployment and the benchmark operate on French text in Latin script with standard diacritics; no script, modality, or infrastructure mismatch exists. The benchmark's HuggingFace toolkit and tokenization analysis [Q48, Q89, Q90] support transferability. The minor concerns are (a) sentence-splitting of long EMEA documents [Q37] which strips structural context relevant to long SmPC/CTD documents, and (b) PxCorpus's transcribed-speech register (PXCORPUS-D11, PXCORPUS-D13) which is incompatible with regulatory written prose. IF is LOWER priority per the elicitation, and these issues do not threaten core form-level validity.

Strengths

- French Latin-script text consistent with deployment environment across all datasets [Q31]
- HuggingFace Datasets toolkit with normalized loaders supports reproducibility [Q48]
- Tokenization study shows segmentation differences have minor impact on performance [Q78, Q79, Q80]
- Hyperparameters fully documented for reproducibility [Q49, Q99]

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: No signal-distribution mismatch — both deployment and benchmark are written French text in Latin script [Q31]. Tokenization variance was studied and found to have minor impact [Q79, Q80].

IF-2: Regional infrastructure (EU pharmaceutical companies, ANSM/EMA) is text-only enterprise document processing, fully compatible with the benchmark's text format [Q48].

IF-3: Two domain-specific form differences: (a) Long-document handling: SmPCs/CTDs exceed transformer input limits; benchmark mitigates via sentence-splitting [Q37], but this strips section/cross-reference context (DEFT2020-D21, QUAERO-D22, QUAERO-D24). (b) PxCorpus transcribed speech (PXCORPUS-D11, PXCORPUS-D13) introduces disfluencies and English fragments not present in regulatory documents.

ID	Evidence
PXCORPUS-D 11	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English fragments, expletives, off-task speech; absent from regulatory text
PXCORPUS-D 13	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment — System error message transcribed as text; impossible in written regulatory documents
DEFT2020-D21	En cas de prise le soir, il faut recommander de ne pas s'aliter dans les 2 heures suivant celle-ci et de tenir compte du délai d'action (6-8 heures). — Posology instruction decontextualized by sentence splitting; section/drug identity lost

IF-4: Form mismatches are minor: sentence-splitting context loss for long regulatory documents, and the PxCorpus speech-transcription register being inapplicable to formal regulatory prose. No script, encoding, or modality issues.

ID	Evidence
QUAERO-D22	`"o ."' — Single-character fragment from sentence splitting
QUAERO-D25	`"Č eská republika Biogen Idec (Czech Republic) s ."' — Czech-language text embedded in French EMEA document

Requires Expert Verification

- Whether sentence-splitting context loss materially impairs SmPC/CTD compliance verification in deployment

Remediation

Gap 1: Sentence-splitting of long EMEA documents [Q37] strips section structure that is legally and semantically critical in SmPC/CTD compliance verification.

Recommendation: When applying DrBenchmark-trained models to deployment SmPC/CTD documents, implement section-aware chunking that preserves QRD section identity (4.1 Indications, 4.2 Posology, 4.3 Contraindications, 4.4 Special warnings) as metadata accompanying each chunk, so downstream regulatory rules can use section context that the benchmark training did not preserve.